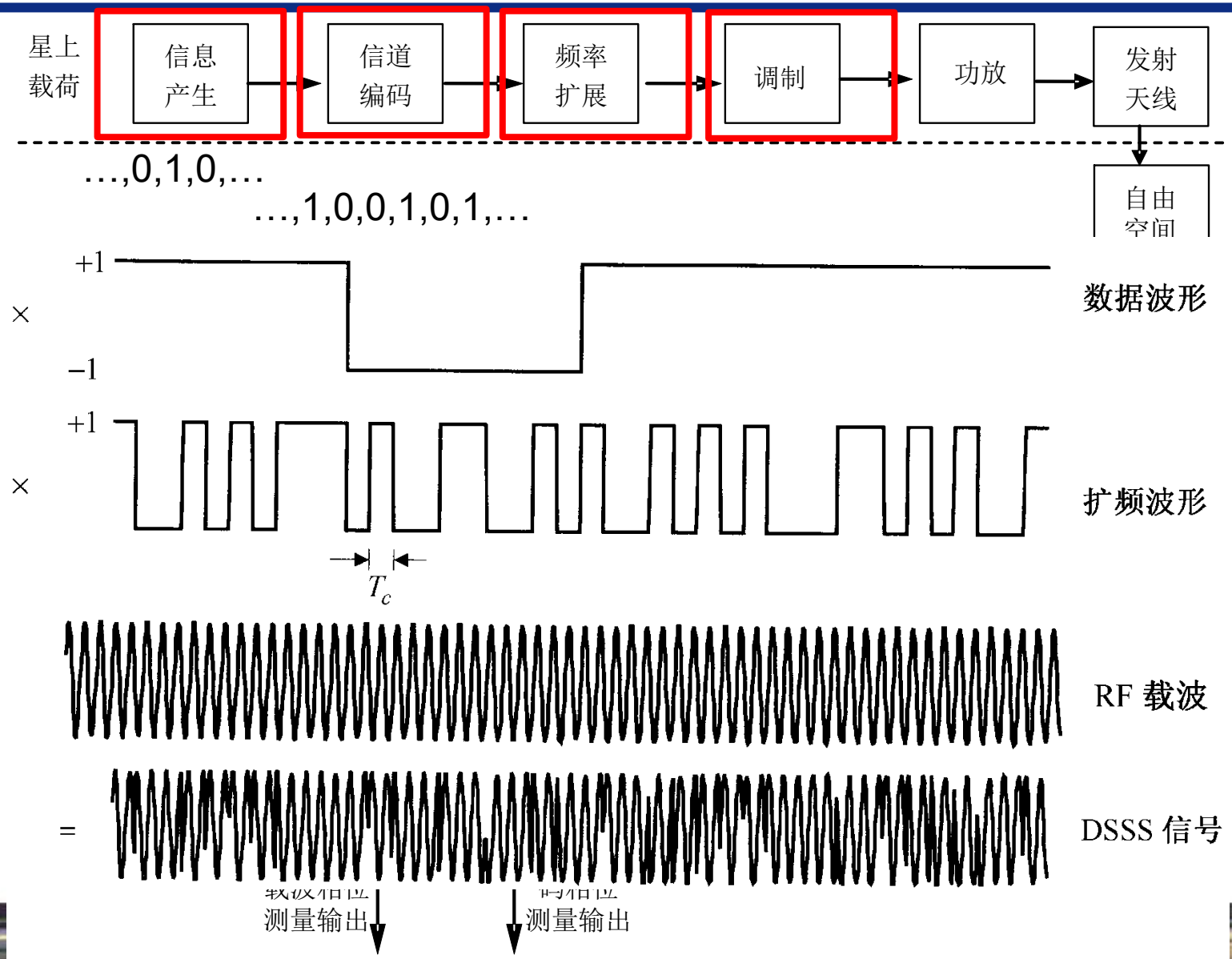
直接序列扩频信号捕获
——原理及仿真

主讲人：唐祖平

主要内容

- 基本原理
- 基于matlab的直扩信号捕获仿真
- 实验内容

基本原理


星上
载荷
信息
产生
信道
编码
频率
扩展
调制
功放
发射
天线
自由
空间
…,0,1,0,…
…,1,0,0,1,0,1,…
+1
–1
数据波形
×
+1
×
扩频波形
T_c
RF 载波
=
DSSS 信号
测量输出
测量输出

基本原理

射频信号的复基带表达

射频信号 $s_{\mathrm{RF}}(t) = s_I(t)cos(2\pi f_c t) - s_Q(t)sin(2\pi f_c t)$

复基带信号表达 $s_B(t) = s_I(t) + js_Q(t)$

$$s_B(t)e^{j2\pi f_c t} = [s_I(t) + js_Q(t)][cos(2\pi f_c t) + jsin(2\pi f_c t)]$$

$$= s_I(t)cos(2\pi f_c t) - s_Q(t)sin(2\pi f_c t) \qquad s_{RF}(t)$$

$$+\mathrm{j}[s_I(t)sin(2\pi f_c t) + s_Q(t)cos(2\pi f_c t)] \qquad js'_{RF}(t)$$

$$s_{RF}(t) = Re\{s_B(t)e^{j2\pi f_c t}\}$$

$$s_B(t) = [s_{RF}(t) + js'_{RF}(t)]e^{-j2\pi f_c t}$$

导航信号模型

考虑多普勒和初始相位

$$s_{RF}(t) = D(t)C(t)cos[2\pi(f_c + f_d)t + \varphi]$$

$$s_B(t) = [s_{RF}(t) + \mathrm{j}s'_{RF}(t)]e^{-j2\pi f_c t}$$

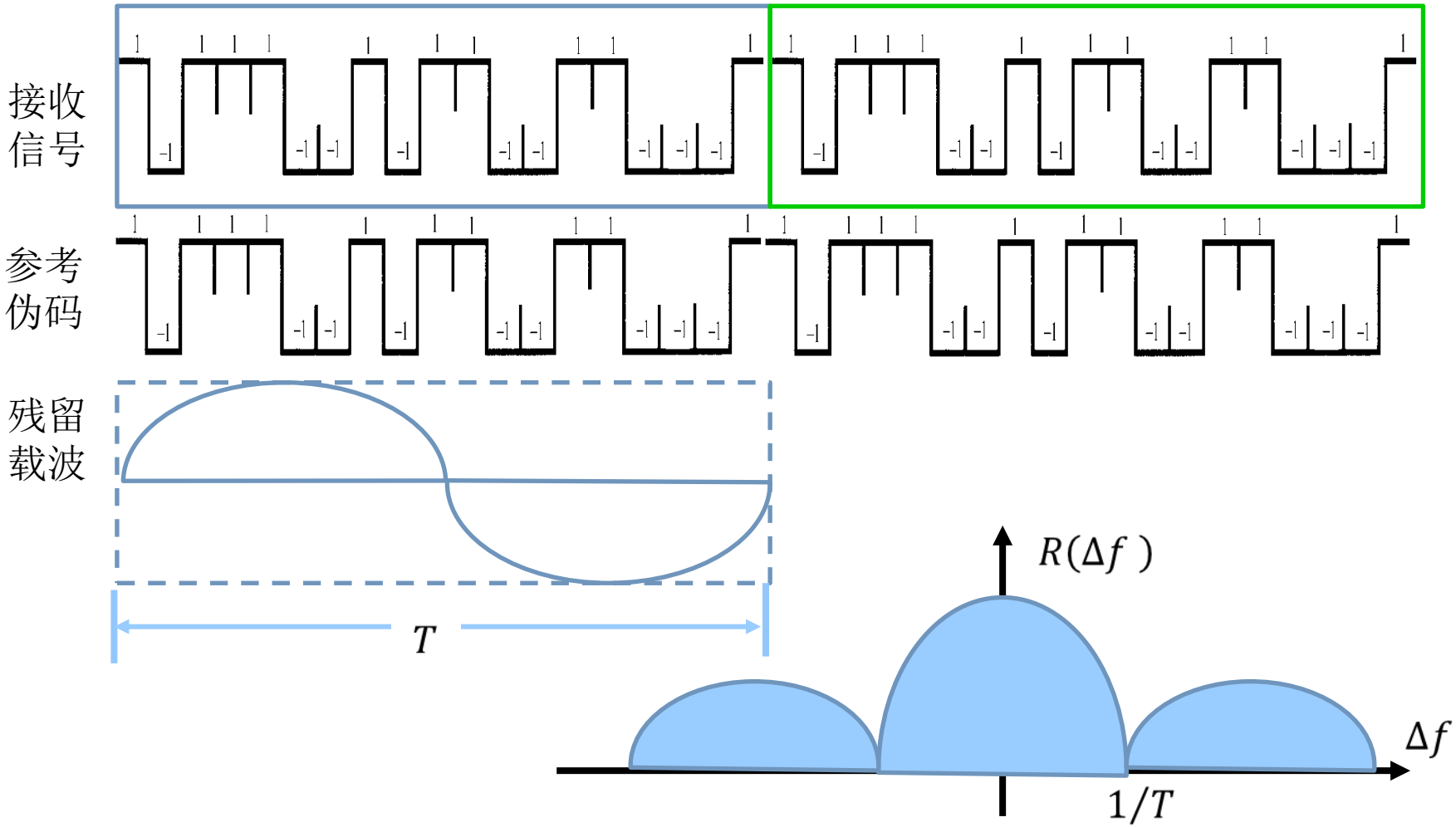
$$= \begin{bmatrix} D(t)C(t)cos[2\pi(f_c + f_d)t + \varphi] \\ +\mathrm{j}D(t)C(t)sin[2\pi(f_c + f_d)t + \varphi] \end{bmatrix} e^{-j2\pi f_c t}$$

$$= D(t)C(t)e^{j(2\pi(f_c+f_d)t + \varphi)}e^{-j2\pi f_c t}$$

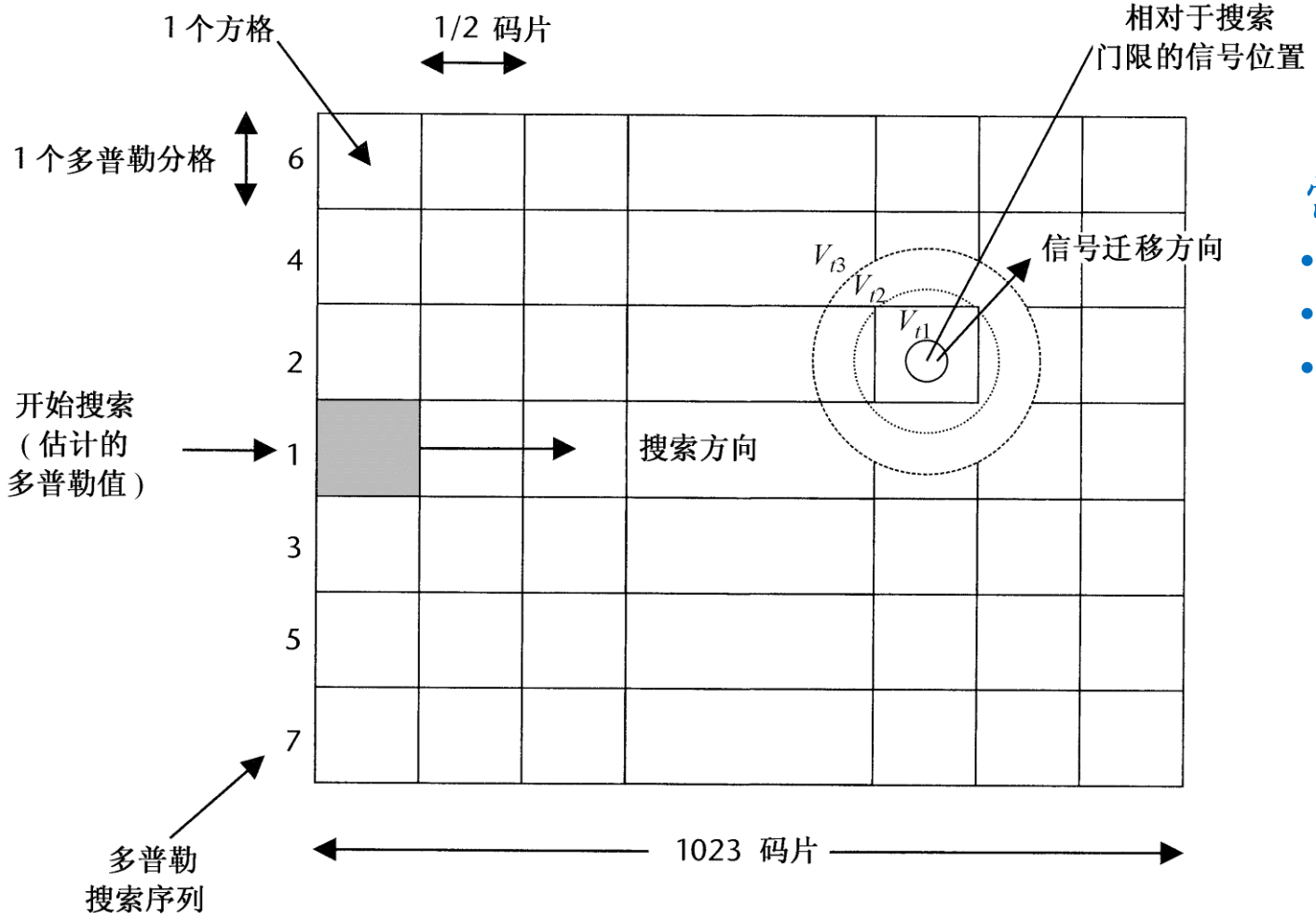
$$= D(t)C(t)e^{j(2\pi f_d t + \varphi)}$$

$$s_I(t) = D(t)C(t)\cos(2\pi f_d t + \varphi)$$
$$s_Q(t) = D(t)C(t)sin(2\pi f_d t + \varphi)$$

工作原理



工作原理

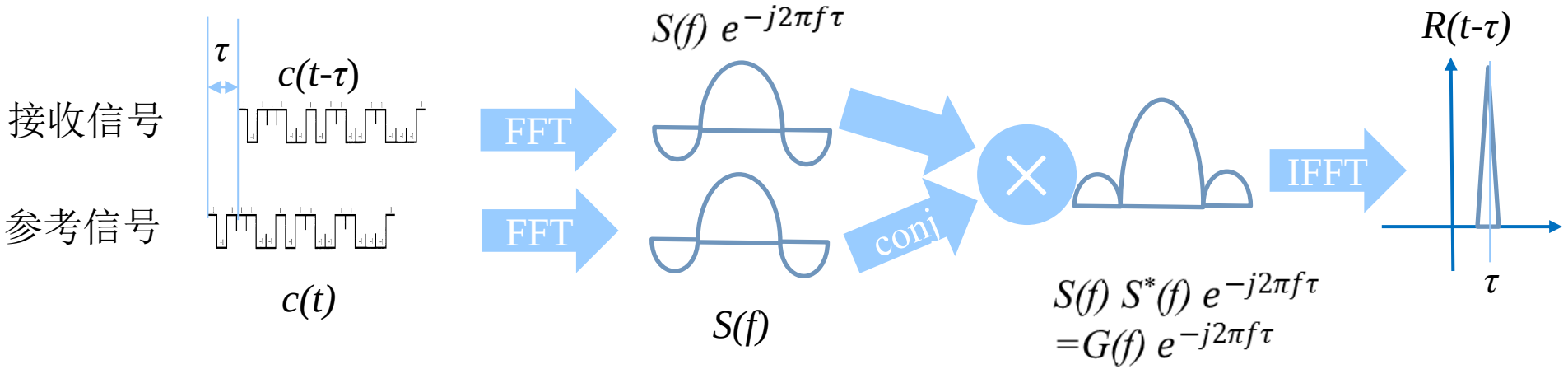


常见捕获方法

- 串行捕获
- 码相位并行捕获
- 载波频率并行捕获

工作原理

❑ 码相位并行捕获

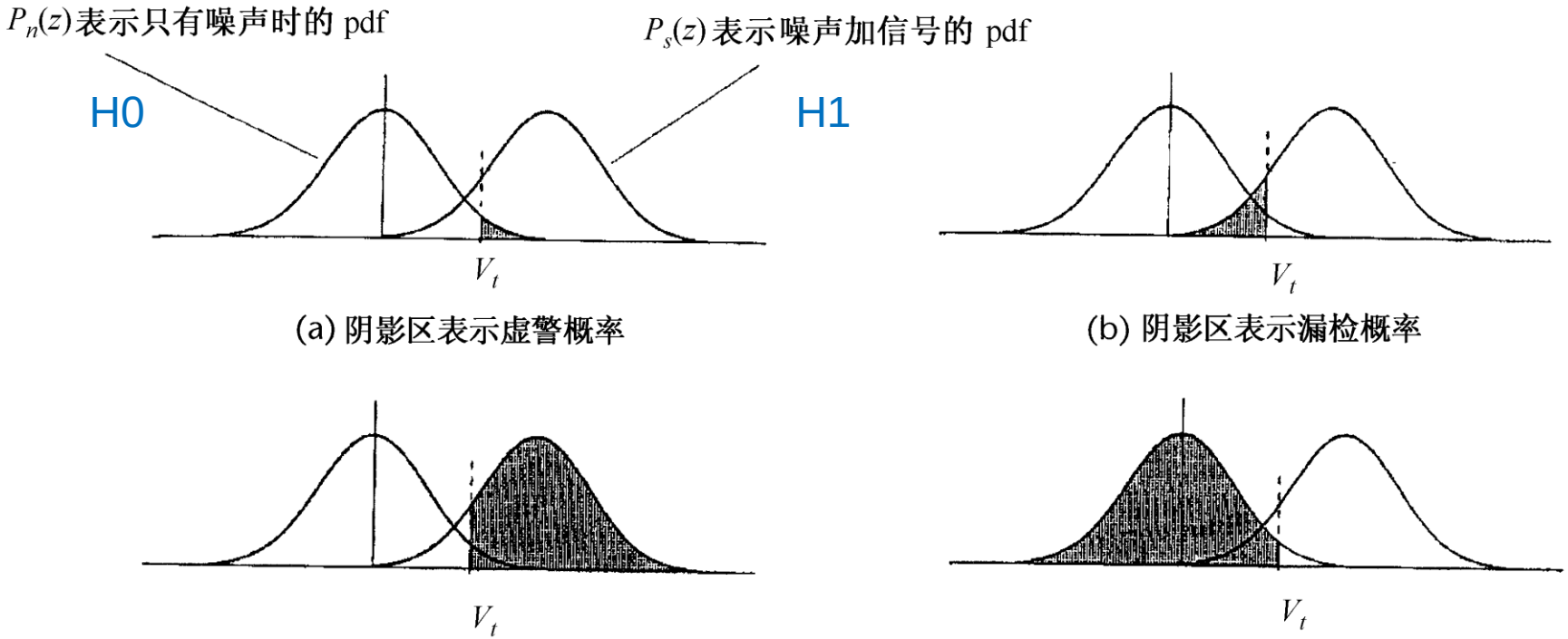


$$\int c(t-\tau)e^{-j2\pi ft}dt = \int c(t-\tau)e^{-j2\pi f(t-\tau+\tau)}d(t-\tau) = e^{-j2\pi f\tau}\int c(t)e^{-j2\pi ft}d(t)$$

工作原理

❑ 虚警、漏检、检测概率

假设检验：
H0：信号不存在
H1：信号存在



(a) 阴影区表示虚警概率

(b) 阴影区表示漏检概率

(c) 阴影区表示检测概率

(d) 阴影区表示正确舍弃

信号捕获仿真

testAcquisition.m

```
%% 参数设置

NumOfPRN =10;                          % PRN号
CAcode = generateCAcode(NumOfPRN);     % 生成GPS C/A 码
Rc  = 1.023e6;                         % 码速率
Tc  = 1/Rc;                            % 码元宽度 s
fs  = 100e6;                           % 采样率
TaoOfRec = 100 * Tc;                   % 接收信号的传播延迟 s
FdOfRec  = 1000;                % 接收信号的多普勒频偏 Hz
Tp  = 1e-3;                            % 积分周期 s
```

信号捕获仿真

- 无噪声条件下的相关特性

testAcquisition.m

```
%% 仿真信号
Tsim    = Tp;                   % 仿真时长
t   = 0:1/fs:Tp;                % 采样时刻

% 接收信号，无噪声
SigRec  = CAcode(mod(floor((t-TaoOfRec)/Tc),CodeLen)+1) .*
exp(1i * 2 *pi* FdOfRec * t);

% 参考信号
SigRef  = CAcode(mod(floor(t/Tc),CodeLen)+1);
```

信号捕获仿真

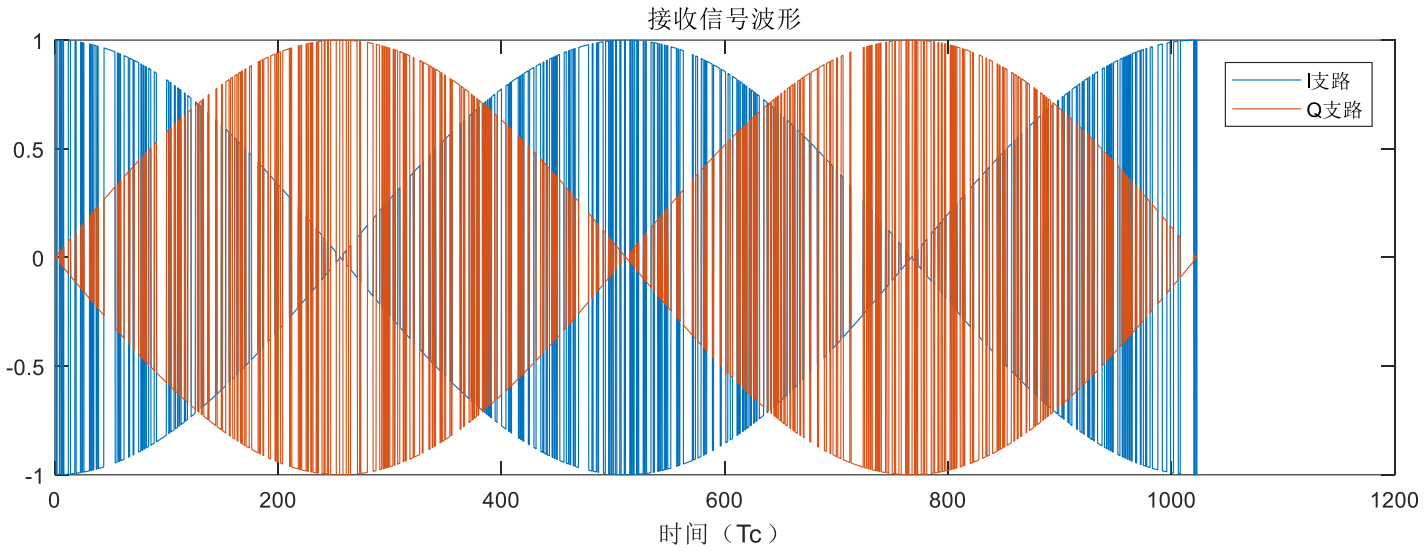
testAcquisition.m

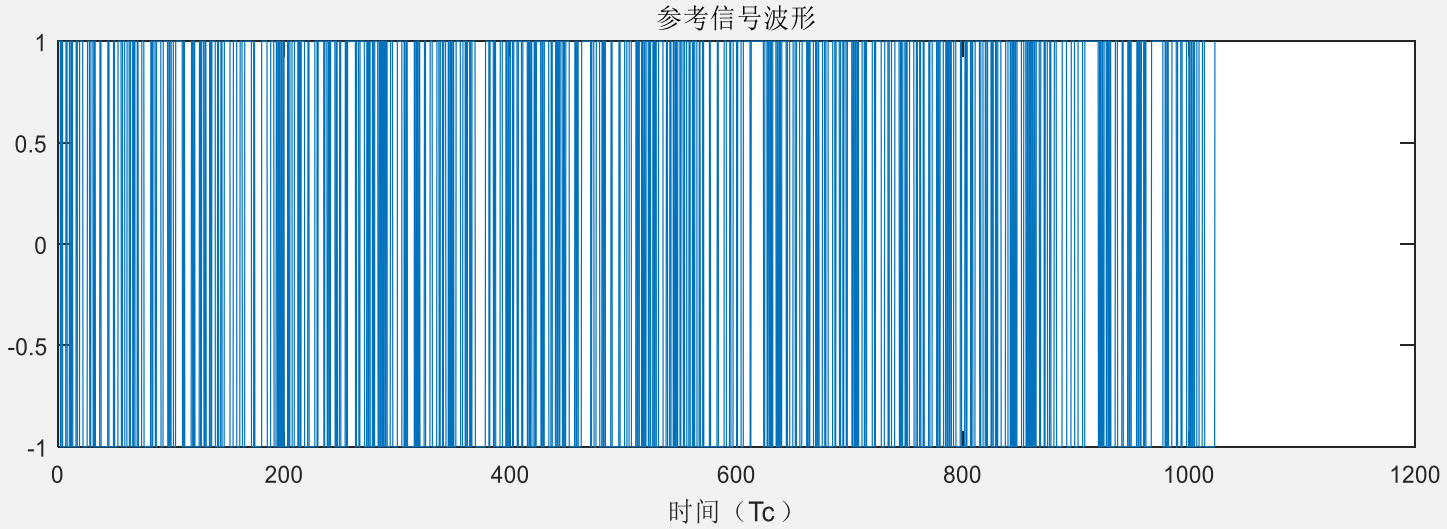
```
%% 仿真相关输出

DopplerTest = -5000:500:5000;                    % 多普勒频偏测试点
ACFs = zeros(length(DopplerTest),length(SigRec));    % 相关值矩阵

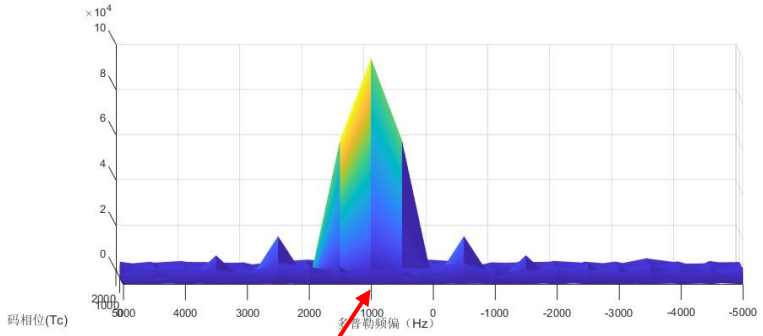

for m = 1:length(DopplerTest)
    FdOfRef = DopplerTest(m);
    % 多普勒消除
    SigRec1 = SigRec .* exp(-1i * 2 *pi* FdOfRef * t);
    % 基于FFT的码相位并行搜索
    ACFs(m,:) = ifft(fft(SigRec1) .* conj(fft(SigRef)));
end
```

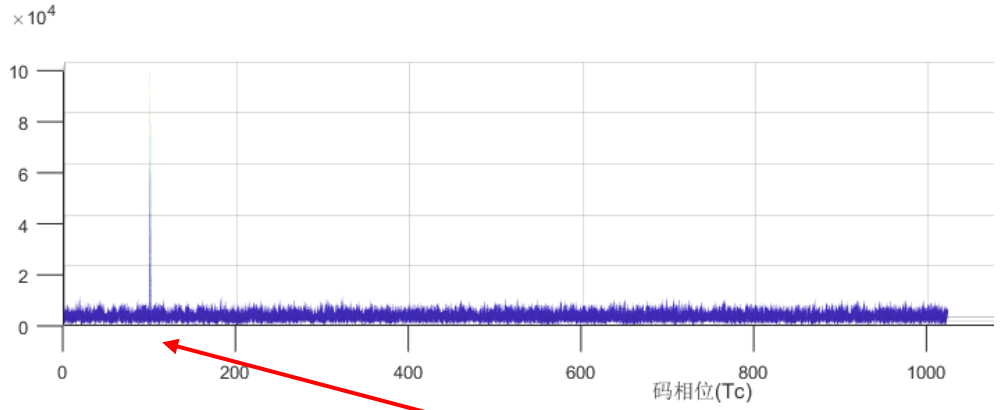
信号捕获仿真


接收信号波形
1
0.5
0
-0.5
-1
0
200
400
600
800
1000
1200
I支路
Q支路
时间（Tc）


参考信号波形
1
0.5
0
-0.5
-1
0
200
400
600
800
1000
1200
时间（Tc）

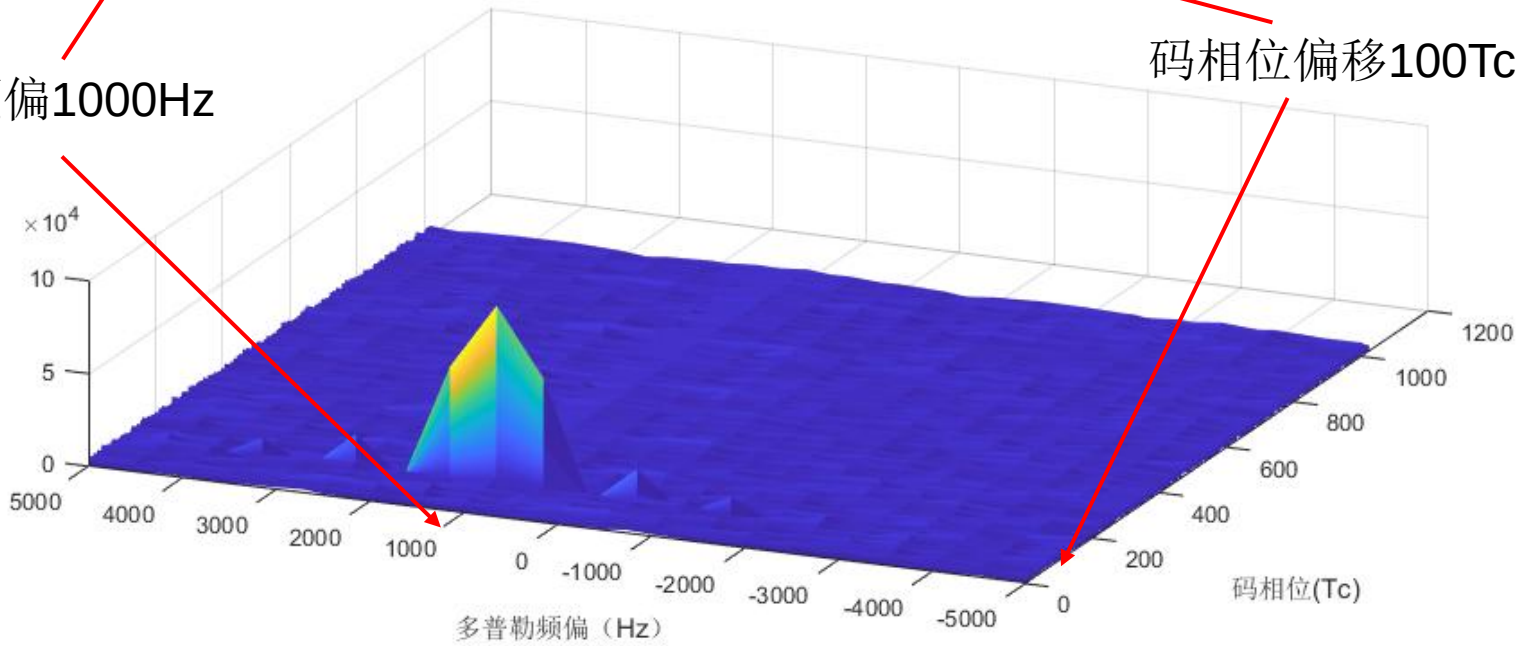
信号捕获仿真





多普勒频偏1000Hz

码相位偏移100Tc



信号捕获仿真

- AWGN条件下的捕获性能

testAcquisitionNoise.m

```
%% 参数设置
NumOfPRN =10;                                   % PRN号
CAcode = generateCAcode(NumOfPRN);              % 生成GPS C/A 码
Rc  = 1.023e6;                                  % 码速率
Tc  = 1/Rc;                                     % 码元宽度 s
fs  = Rc;                                       % 采样率
TaoOfRec = 0 * Tc;                              % 接收信号的传播延迟 s
FdOfRec  = 0;                                   % 接收信号的多普勒频偏 Hz
CNRdBs   = 35;                                  % 接收信号载噪比
SNR      = (10.^(CNRdBs/10)/fs);                % 接收信号信噪比
Tp  = 1e-3;                                     % 积分周期 s
```

信号捕获仿真

- AWGN条件下的捕获性能

testAcquisitionNoise.m

```
%% 仿真信号
Tsim    = Tp;                   % 仿真时长
t   = 1/fs:1/fs:Tp;             % 信号采样时刻
LenSig = length(t);             % 采样点数
% 接收信号，加噪声
SigRec  = sqrt(2*SNR)*CAcode(mod(floor((t-TaoOfRec)/Tc),CodeLen)+1)
          + randn(1,LenSig)+ 1i*randn(1,LenSig);
% 参考信号
SigRef  = CAcode(mod(floor(t/Tc),CodeLen)+1);
```

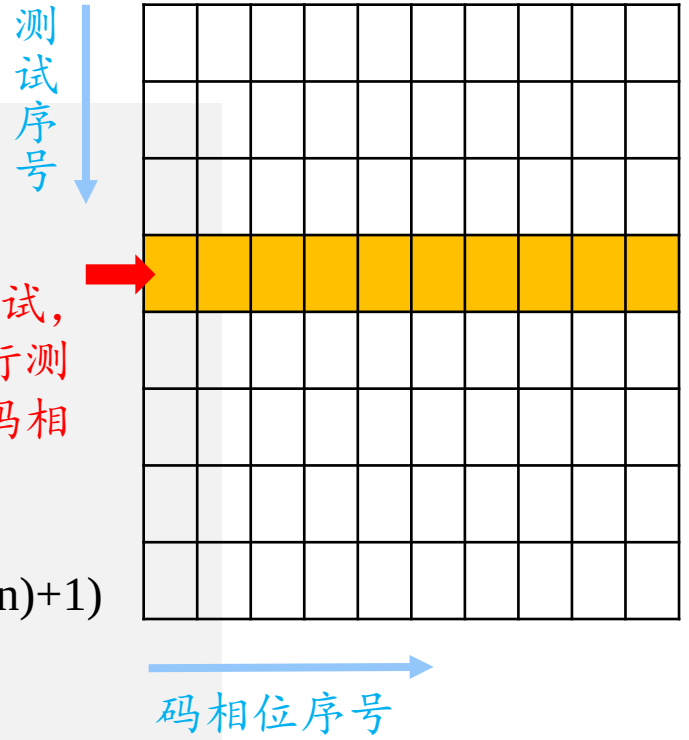
信号捕获仿真

- AWGN条件下的捕获性能

testAcquisitionNoise.m

```
%% 捕获测试
Ntest = 10000;                          % 并行捕获次数
ACFs = zeros(Ntest, length(SigRec));    % 相关值存储矩阵


for m = 1:Ntest
    SigRec  = sqrt(2*SNR)*CAcode(mod(floor((t-TaoOfRec)/Tc),CodeLen)+1)
              + randn(1,LenSig)+ 1i*randn(1,LenSig);
    ACFs(m,:) = abs(ifft(fft(SigRec) .* conj(fft(SigRef))));
    waitbar(m/Ntest);
end
```

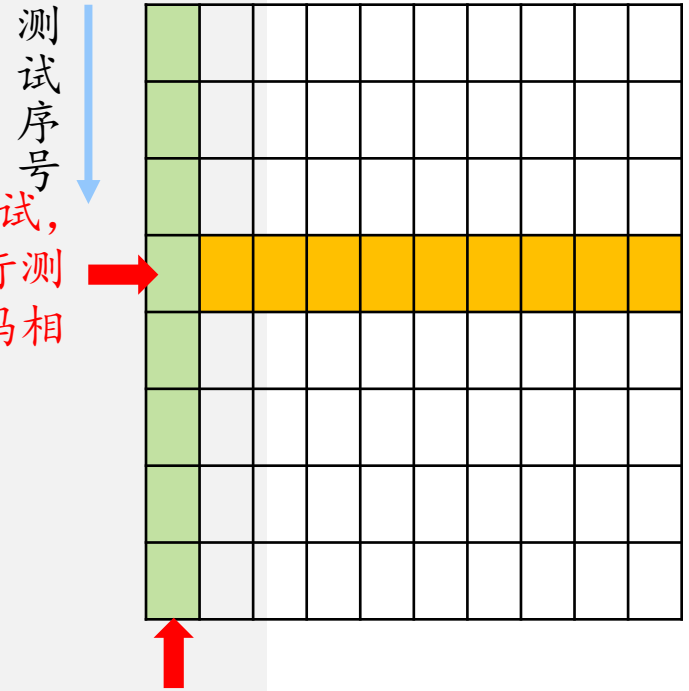


信号捕获仿真

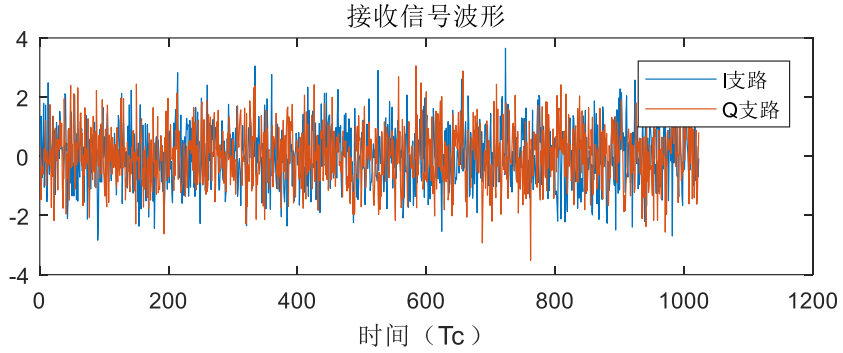
- AWGN条件下的捕获性能

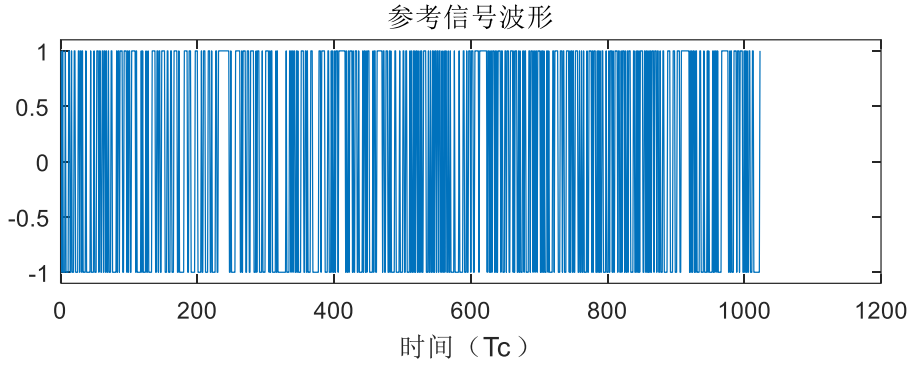
testAcquisitionNoise.m

```
%% 统计检测概率
THs  = 0.1:0.1:5;                    % 判决门限(×噪声标准差)
Pd = zeros(size(THs));               % 检测概率
Pf = zeros(size(THs));               % 虚警概率
for m = 1:length(THs)
    Th = THs(m) * sqrt(LenSig);
    % H1假设检验
    Pd(m) = nnz(ACFs(:,1)>=Th)/Ntest;
    % H0假设检验
    Pf(m) = nnz(ACFs(:,2:LenSig)>=Th)/(Ntest*(LenSig-1));
end
```

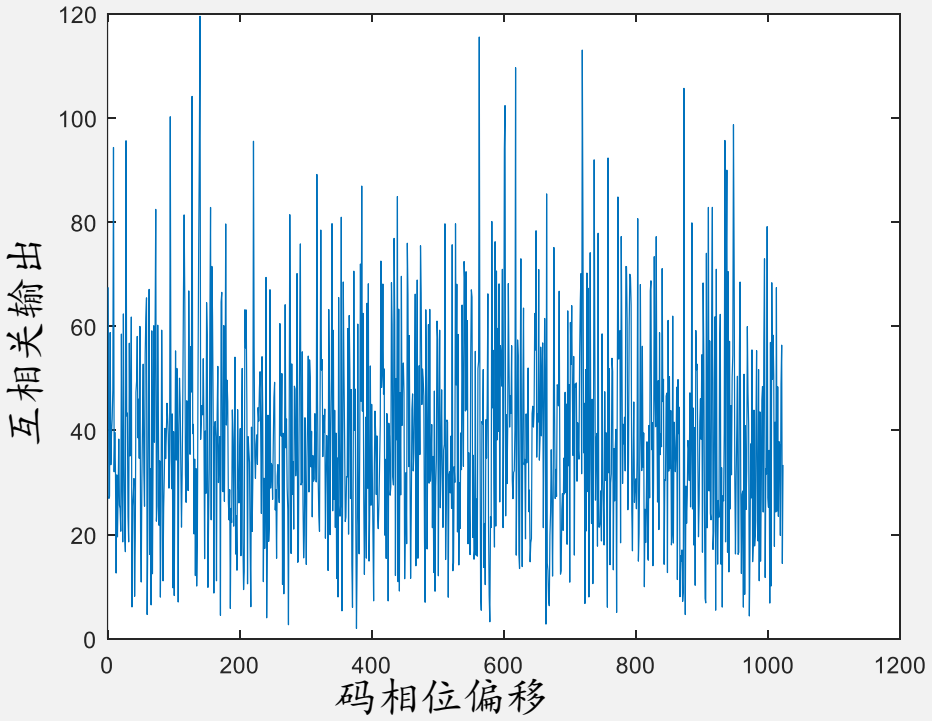


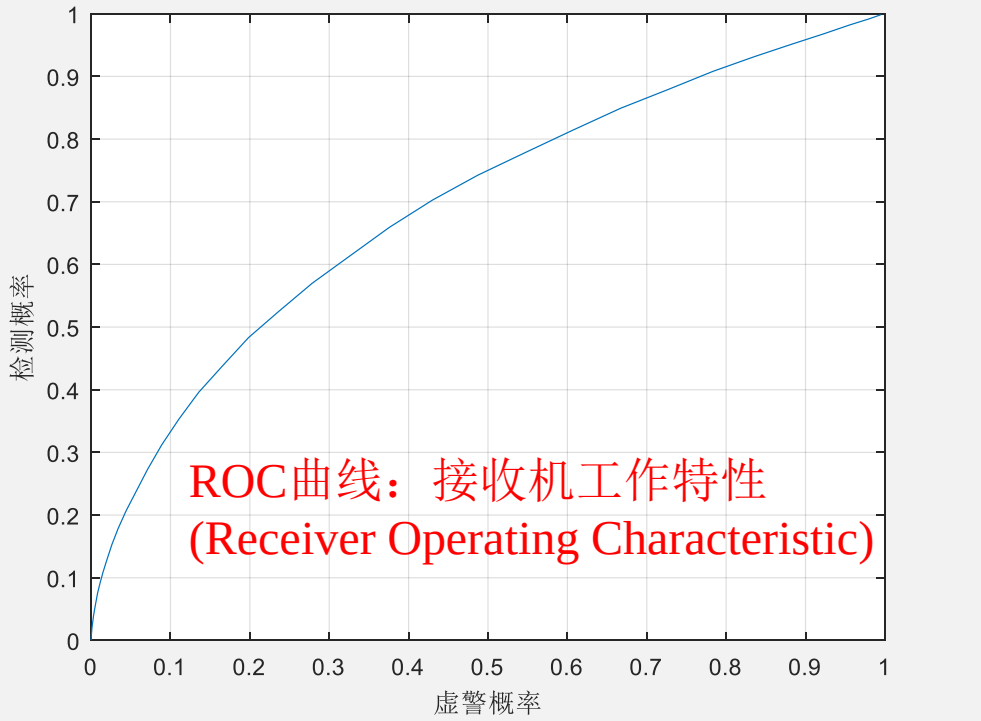
信号捕获仿真



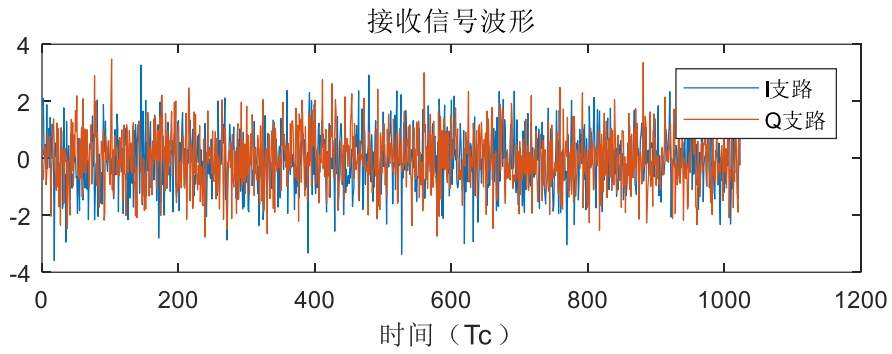


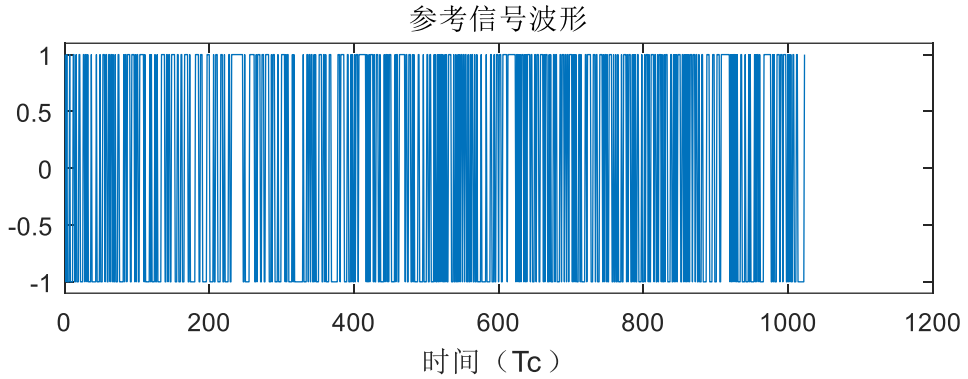
载噪比30dBHz, 积分时间1ms



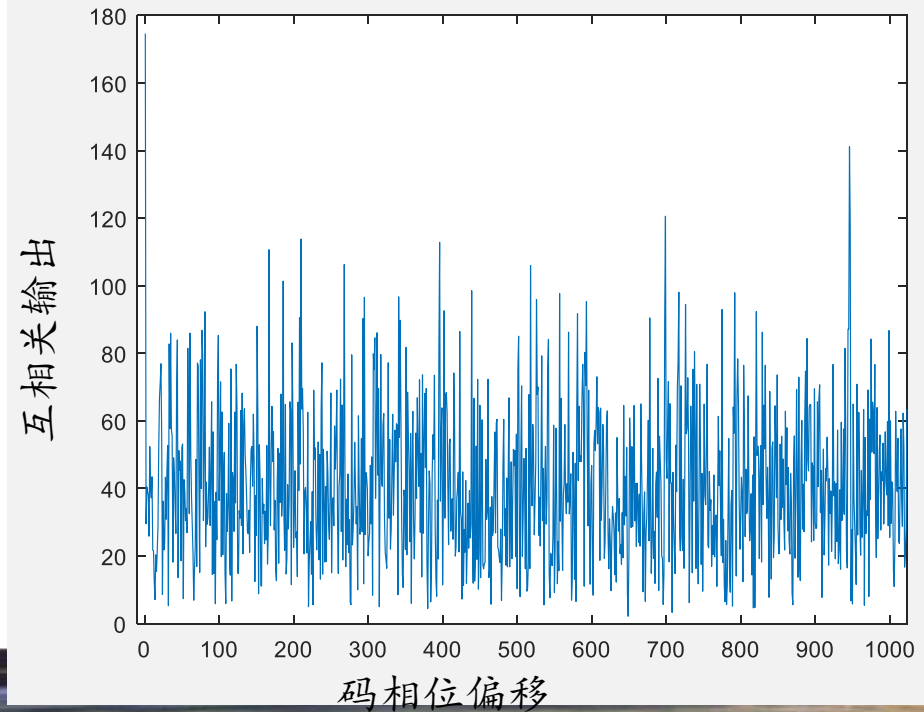


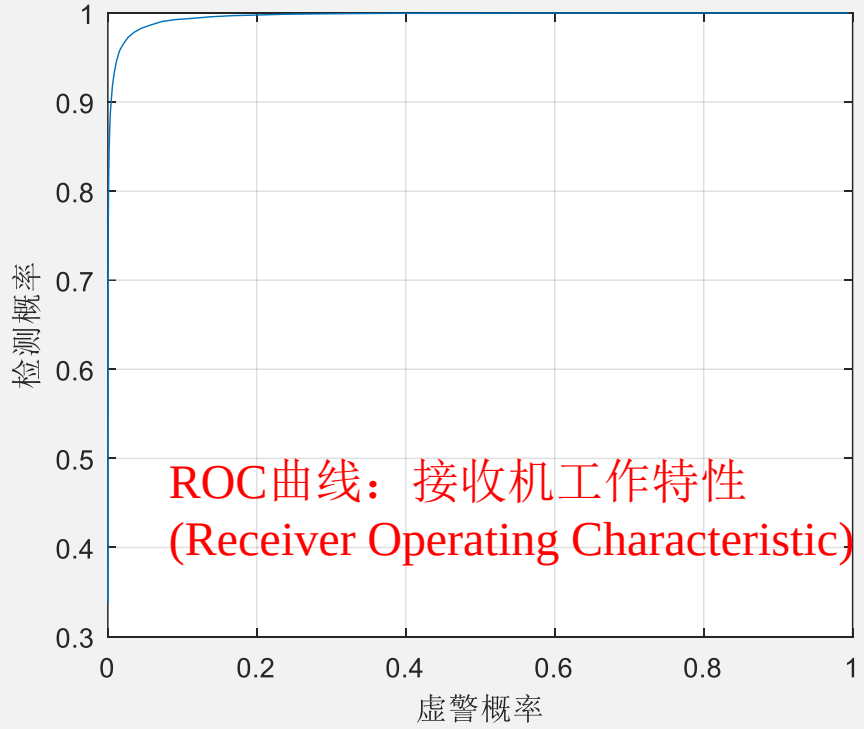
信号捕获仿真



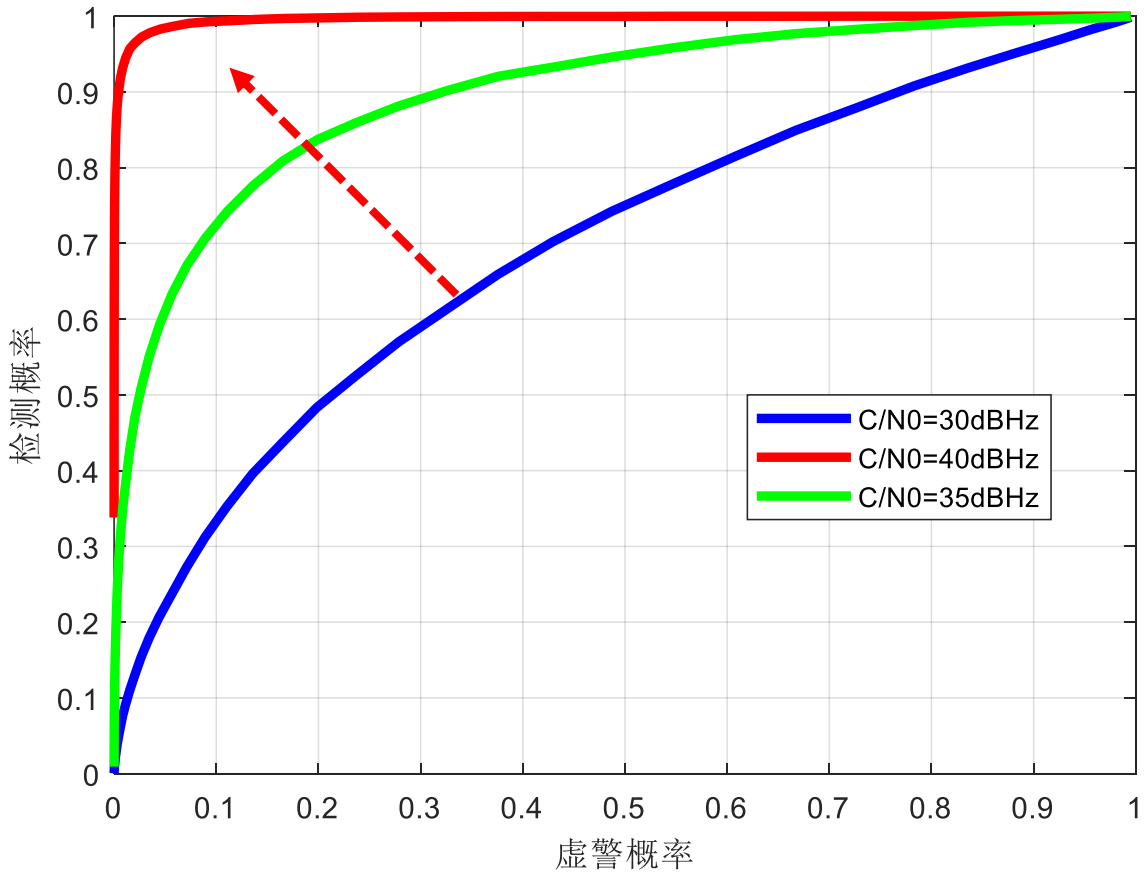


载噪比40dBHz, 积分时间1ms





信号捕获仿真



曲线向箭头方向移动，接收机性能变好or坏？

ROC曲线：接收机工作特性
(Receiver Operating Characteristic)

实验内容

- 相关曲面仿真
  - AWGN，C/N0=40dBHz
  - 多普勒偏移、码延迟自定
- 接收机工作特性曲线仿真（Pd vs. Pf）
  - C/N0=30,35,40 (dBHz)
- 检测概率（ Pd vs. C/N0）仿真
  - 判决门限=1σ、2σ、3σ
  - C/N0 = 25:1:45 (dBHz)
- 虚警概率（ Pd vs. C/N0）仿真
  - 判决门限=1σ、2σ、3σ
  - C/N0 = 25:1:45 (dBHz)